

Job Posting System

Project Documentation Report

Programming Language: Python

GUI Framework: PySide6

Database: SQLite

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2026

1. Project Objectives

The main goal of this project is to develop a desktop-based job posting system where employers can post job listings and job seekers can browse and apply for those listings. The system is designed to support multiple user types with different roles and permissions.

The specific objectives of the project are:

- Allow employers to create, edit, and manage their job postings
- Allow job seekers to search for jobs, view details, and submit applications
- Provide an admin panel to manage all users and job listings from a central point
- Generate summary reports for the admin to monitor system activity
- Store all data persistently using a relational database

2. Technologies Used

The following technologies and libraries were used in this project:

Component	Technology	Purpose
Programming Language	Python 3.10	Core application logic
GUI Framework	PySide6	Desktop user interface
Database	SQLite	Data storage
UI Design	Qt Designer (.ui files)	Visual layout design
IDE	PyCharm	Development environment

3. System Architecture

The project follows a simple layered architecture. The presentation layer consists of PySide6 screens and widgets. Below that, a business logic layer handles authentication and report generation. At the bottom, a data access layer communicates with the SQLite database through a DatabaseManager class.

The project is organized into the following main components:

File / Module	Role
main.py	Entry point, starts the Qt application
database.py	DatabaseManager class, handles all SQL operations
auth_manager.py	Handles login, registration, and session management
user.py	User model class
job.py	Job and JobCategory model classes

employer.py	Employer profile and job management
job_seeker.py	Job seeker profile and application retrieval
application.py	Application model and status management
report_generator.py	Generates statistics for the admin panel
main_window.py	Main window that switches between screens
login_screen.py	Login screen logic
register_screen.py	Registration screen logic
job_listing_screen.py	Job browsing screen for job seekers
job_detail_screen.py	Detailed job view and application submission
employer_dashboard.py	Employer dashboard for managing listings
admin_panel.py	Admin panel for user/job management and reports
profile_screen.py	User profile editing screen
search_filter_widget.py	Search and filter widget for job listings

4. Database Design

The application uses SQLite as the database engine. The database file is created automatically when the application starts. It contains the following tables:

4.1 users

Stores all user accounts regardless of role.

Column	Type	Description
id	INTEGER PK	Auto-incremented primary key
username	TEXT UNIQUE	Login username
password	TEXT	User password (plain text)
email	TEXT UNIQUE	User email address
role	TEXT	One of: jobseeker, employer, admin
created_at	TEXT	Account creation timestamp

4.2 jobs

Stores all job postings created by employers.

Column	Type	Description
id	INTEGER PK	Auto-incremented primary key
employer_id	INTEGER FK	References users.id
category_id	INTEGER FK	References job_categories.id
title	TEXT	Job title

company	TEXT	Company name
location	TEXT	Job location
salary	TEXT	Salary info (optional)
description	TEXT	Job description
is_active	INTEGER	1 = active, 0 = passive
created_at	TEXT	Posting timestamp

4.3 applications

Stores job applications submitted by job seekers.

Column	Type	Description
id	INTEGER PK	Auto-incremented primary key
job_id	INTEGER FK	References jobs.id
applicant_id	INTEGER FK	References users.id
cover_letter	TEXT	Optional cover letter text
status	TEXT	pending / accepted / rejected
applied_at	TEXT	Application timestamp

4.4 profiles

Stores extended profile information for users.

Column	Type	Description
id	INTEGER PK	Auto-incremented primary key
user_id	INTEGER UNIQUE FK	References users.id
full_name	TEXT	Full name of the user
phone	TEXT	Phone number
bio	TEXT	Short biography
skills	TEXT	Skills as a comma-separated string

4.5 job_categories

Stores predefined job categories. Default categories are: Software, Marketing, Design, Accounting, Human Resources, Sales, Other. New categories can be added by the admin.

5. System Functionalities

5.1 Authentication

Users can register with a username, email, password, and role selection (job seeker or employer). The system validates input fields, checks for duplicate usernames or emails, and

shows appropriate error messages. Login is handled by the AuthManager class which stores the current session. The admin account is seeded automatically into the database on first run with the credentials admin / admin123.

5.2 Job Seeker Features

- View a list of all active job postings
- Search jobs by keyword, category, or location using the search and filter widget
- Click on a job card to view the full job details
- Submit an application with an optional cover letter
- The system prevents duplicate applications to the same listing
- View all submitted applications and their current statuses (pending, accepted, rejected)
- Edit personal profile including full name, phone, bio, and skills

5.3 Employer Features

- Post new job listings through a dialog form with fields for title, company, location, salary, category, and description
- View all posted jobs in a dashboard table
- Edit existing job listings
- Toggle a job listing between active and passive state
- Delete a job listing
- View applicants for each job listing
- Accept or reject individual applications

5.4 Admin Panel

- View a table of all registered users with their ID, username, email, role, and registration date
- Delete any user from the system
- View a table of all job listings with status and creation date
- Delete any job listing
- View a summary report section showing total users, total listings, active listings, and total applications

5.5 Search and Filtering

The job listing screen includes a search and filter widget. Users can enter a keyword to search across job title, company name, and description. They can also filter by job category using a dropdown and by location using a text field. The search is performed with a SQL LIKE query on the database.

5.6 Report Generation

The ReportGenerator class provides statistical queries for the admin panel. It returns the total number of users, total and active job listings, total number of applications, a breakdown of applications by status, the top five most applied jobs, and the number of listings per category.

6. User Roles Summary

Feature	Job Seeker	Employer	Admin
Register / Login	Yes	Yes	Yes (seeded)
Browse job listings	Yes	No	No
Apply to jobs	Yes	No	No
View own applications	Yes	No	No
Post job listings	No	Yes	No
Manage own listings	No	Yes	No
View applicants	No	Yes	No
Accept/Reject applications	No	Yes	No
Edit profile	Yes	Yes	No
Manage all users	No	No	Yes
Manage all listings	No	No	Yes
View system reports	No	No	Yes

7. Conclusion

This project implements a functional job posting system as a desktop application using Python and PySide6. The system covers the basic workflow of job posting and application management with three distinct user roles. The database design uses foreign keys and referential integrity to ensure data consistency. The UI is built using Qt Designer and connected to the backend through PySide6 bindings.

The project could be extended in the future with features such as password hashing for security, file upload support for resumes, or a web-based version using a framework like Django or FastAPI.